

4.4.2 Ecosystems and biodiversity

Ecosystems with high levels of biodiversity help to provide the resources needed to sustain life on Earth, including human life. This makes it very important that scientists understand the relationships within and between communities of organisms. The science helps to evaluate the negative and positive human impacts on biodiversity of human activities both locally and globally.

The required practical is an investigation of factors affecting population size of a common species in a habitat.

4.4.2.1 Levels of organisation in an ecosystem

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe different levels of organisation in an ecosystem from individual organisms to the whole ecosystem.	An ecosystem is made up of all the living organisms in a particular environment together with the non-living components such as soil, air and water. A habitat is where a particular organism lives in an ecosystem. A population is made up of all the individuals of the same species in a habitat. A community is made up of all the populations of different organisms that live in the same habitat. Feeding relationships within a community can be represented by food chains. All food chains begin with a producer that synthesises molecules. This is usually a green plant, which absorbs light to make glucose. A food web can be used to understand the interdependence of species within an ecosystem in terms of food sources. Producers are eaten by primary consumers, which in turn may be eaten by secondary consumers and then tertiary consumers. Consumers that kill and eat other animals are predators, and those eaten are prey. In a community the numbers of predators and prey rise and fall in cycles.	WS 1.2 Interpret graphs used to model predator–prey cycles.

4.4.2.2 Interdependence and competition

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the importance of interdependence and competition in a community.	To survive and reproduce, organisms require a supply of materials from their surroundings and from the other living organisms in an ecosystem. Plants often compete with each other for light and space, and for water and nutrients from the soil. Animals often compete with each other for food, mates and territory. Within a community each species depends on other species for food, shelter, pollination, seed dispersal etc. If one species is removed it affects the whole community. A stable community is one where all the species and environmental factors are in balance so that population sizes remain fairly constant.	

4.4.2.3 Factors that affect communities

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how some abiotic and biotic factors affect communities.	Abiotic factors that can affect a community are: • light intensity • temperature • moisture levels • soil pH and mineral content • wind intensity and direction • carbon dioxide levels for plants • oxygen levels for aquatic animals. Biotic factors that can affect a community are: • availability of food • new predators arriving • new diseases • one species outcompeting another.	WS 1.2 Predict how a change in an abiotic, or biotic, factor would affect a given community given appropriate data or context. MS 1c Calculate the percentage of mass. MS 2c, 4a Extract and interpret information from charts, graphs and tables.

4.4.2.4 Field investigations

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe how to carry out a field investigation into the distribution and abundance of organisms in an ecosystem and explain how to determine their numbers in a given area.	Ecologists use a range of investigation methods using transects and quadrats to determine the distribution and abundance of species in an ecosystem.	MS 2b Calculate arithmetic means. WS 3.3 Carry out and represent mathematical and statistical analysis. MS 4a, 4c Plot and draw appropriate graphs, selecting appropriate scales for the axes. MS 2d Understand the principles of sampling as applied to scientific data.

Required practical activity 12: measure the population size of a common species in a habitat. Use sampling techniques to investigate the effect of a factor on the distribution of this species.

AT skills covered by this practical activity: biology AT 1, 3, 4 and 6.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 169).

4.4.2.5 Biodiversity

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain some of the benefits and challenges of maintaining local and global biodiversity.	Biodiversity is greater in ecosystems that provide a bigger range of different habitats, which are home to larger populations of a variety of organisms. Small populations are in greater danger of dying out if an ecosystem is disrupted in some way. Ecosystems with high levels of biodiversity help to provide the resources needed to sustain life, including human life. Ecosystems with higher biodiversity offer economic benefits by sustaining the resources needed for agriculture, fishing and forestry.	

4.4.2.6 Negative human impacts on ecosystems

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe negative human interactions within ecosystems and explain their impact on biodiversity.	Examples of human interactions with local ecosystems that can diminish or destroy biodiversity include: • building, quarrying, farming, clearing woods and other activities that destroy habitats • the destruction of peat bogs, and other areas of peat, to produce garden compost • pollution of streams, rivers and lakes by sewage, toxic wastes and fertilisers. An example of a global impact of human activities is global warming leading to climate change (The Earth's atmosphere (page 67)).	WS 1.4 Evaluate given information about ways in which human activities affect the environment.

4.4.2.7 Positive human impacts on ecosystems

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe positive human interactions within ecosystems and explain their impact on biodiversity.	There are programmes to reduce these negative effects on ecosystems and biodiversity. These include: • breeding programmes for endangered species • protecting and regenerating habitats • reintroducing wider field margins and hedgerows in areas of monoculture • recycling resources rather than dumping waste in landfill • production of peat-free composts • reducing deforestation and carbon dioxide emissions.	WS 1.4 Evaluate given information about methods that can be used to tackle problems caused by human impacts on the environment.